

ARFAT NAIK

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PROFESSIONAL SUMMARY

Results-driven Data Science and Machine Learning student with hands-on experience building, training, and deploying end-to-end ML systems on real-world datasets. Proficient in Python, Scikit-learn, TensorFlow, Pandas, Flask, and Docker. Experienced in supervised learning, feature engineering, NLP-based recommendation systems, automated ML pipelines, and containerized deployments on cloud platforms. Seeking a Data Science or Machine Learning internship to apply analytical and modeling skills to impactful problems.

TECHNICAL SKILLS

Languages: Python (NumPy, Pandas, Scikit-learn, TensorFlow, Joblib), SQL, JavaScript

Machine Learning: Supervised Learning (Regression, Classification), Feature Engineering, Model Training & Evaluation, Cross-Validation, Hyperparameter Tuning, Class Imbalance Handling

Performance Metrics: ROC-AUC, Precision, Recall, F1-Score, R^2 , RMSE

NLP & Recommendation: TF-IDF Vectorization, Cosine Similarity, Content-Based Filtering, Sparse Matrix Optimization

Data Science: EDA, Data Visualization (Matplotlib, Seaborn, Chart.js), Statistical Analysis, Data Cleaning, Outlier Detection, Missing Value Imputation

MLOps & Tools: Flask (Model Serving), GitHub Actions (CI/CD), Redis, PostgreSQL (Supabase), Git, Jupyter Notebook, VS Code, Docker

PROJECTS

- AtmosIQ - Air Quality Forecasting System** | *Python, Flask, Redis, Supabase, GitHub Actions* Live | GitHub
- Developed an end-to-end **time-series forecasting system** using a **Random Forest regression model** to predict AQI 3 hours ahead, attaining $R^2 \approx 0.85$ through engineered lag features, rolling window statistics, and meteorological variables.
 - Built a fully automated **hourly ML inference pipeline** with GitHub Actions, Redis (real-time caching), and Supabase (PostgreSQL), enabling continuous data ingestion, feature transformation, prediction, and storage with zero manual steps.
 - Deployed a **production Flask dashboard** on Railway with Chart.js-powered visualizations for PM2.5, PM10, current AQI, and 3-hour forecasts, translating ML outputs into actionable environmental insights.

- Credit Risk Prediction System** | *Python, Logistic Regression, Scikit-learn, Flask, Docker* Live | GitHub
- Designed and trained a **Logistic Regression credit risk model** with **cost-sensitive thresholding** to maximize recall on loan default detection while minimizing false approvals.
 - Applied rigorous **data preprocessing**: missing value imputation, outlier handling, feature engineering, train-test leakage prevention, and **class imbalance management** for robust and unbiased model evaluation.
 - Containerized the **Flask REST API** using **Docker** and deployed on **Railway**, providing real-time default probability scores and binary decisions through a live web interface.

- Movie Recommendation System** | *Python, TF-IDF, Cosine Similarity, Flask, Docker, NLP* Live | GitHub
- Built a **content-based NLP recommendation engine** for 5,000+ TMDB movies, using **TF-IDF vectorization and cosine similarity** on a multi-field feature corpus (genres, keywords, cast, director, overview).
 - Optimized similarity computation via **sparse matrix representations**, reducing runtime by **60%** and containerized the **Flask API** using **Docker**, deployed on **Railway** for real-time recommendations with input validation.

EDUCATION

- Rizvi College of Engineering** Mumbai, India
- Bachelor of Engineering in Electronics and Computer Engineering* 2023 – 2027
- CGPA: 7.27/10 | Semester 6
 - Relevant Coursework: Machine Learning, Statistics, Data Structures & Algorithms, Database Management Systems, Linear Algebra

CERTIFICATIONS

- Programming, Data Structures and Algorithms using Python** | NPTEL Certificate